

Practice Problem 1

$$I = \int \left(\log(\log x) + \frac{1}{(\log x)^2} \right) dx$$

Solution

Let

$$I = I_1 + I_2$$

where

$$I_1 = \int \log(\log x) dx, \quad I_2 = \int \frac{dx}{(\log x)^2}$$

Using integration by parts,

$$\int u dv = uv - \int v du$$

For I_1 ,

$$u = \log(\log x), \quad dv = dx$$

$$du = \frac{dx}{x \log x}, \quad v = x$$

Hence,

$$I_1 = x \log(\log x) - \int \frac{dx}{\log x}$$

Also,

$$\frac{d}{dx} \left(\frac{x}{\log x} \right) = \frac{1}{\log x} - \frac{1}{(\log x)^2}$$

Therefore,

$$I_2 = \int \frac{dx}{\log x} - \frac{x}{\log x}$$

Adding,

$$I = x \log(\log x) - \frac{x}{\log x} + C$$

Final Answer

$$\int \left(\log(\log x) + \frac{1}{(\log x)^2} \right) dx = x \log(\log x) - \frac{x}{\log x} + C$$

Practice Problem 2

Evaluate:

$$I = \int_0^{\frac{\pi}{2}} e^x \sin x \, dx$$

Solution (Integration by Parts)

Using,

$$\int u \, dv = uv - \int v \, du$$

Take,

$$u = \sin x, \quad dv = e^x dx$$

Then,

$$du = \cos x \, dx, \quad v = e^x$$

Therefore,

$$I = [e^x \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^x \cos x \, dx$$

Let,

$$J = \int_0^{\frac{\pi}{2}} e^x \cos x \, dx$$

Again by parts,

$$J = [e^x \cos x]_0^{\frac{\pi}{2}} + I$$

Hence,

$$2I = [e^x (\sin x - \cos x)]_0^{\frac{\pi}{2}}$$

$$2I = e^{\frac{\pi}{2}} + 1$$

$$I = \frac{e^{\frac{\pi}{2}} + 1}{2}$$

Final Answer

$$\int_0^{\frac{\pi}{2}} e^x \sin x \, dx = \frac{e^{\frac{\pi}{2}} + 1}{2}$$